



Department of
Electrical & Electronics Engineering
Abdullah Gül University

Project Report

EE404 Antennas – Dipole Array Antenna Design

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1. OBJECTIVE

The objective of this project is to study linear antenna arrays and to characterise the radiation behaviour of a uniform linear array formed by N identical dipole elements. The number of elements is fixed by the project rule to $N = 4$, which corresponds to the largest final digit among the group members' student identification numbers (taken from 2211011004). The array factor of this configuration is first formulated analytically as a function of the element number N , the inter-element spacing d , the progressive phase difference Φ between consecutive elements, the operating frequency f and the wavenumber k . A MATLAB program is then developed so that, for a given element pattern, spacing, phase shift and frequency, it computes the array factor and the resulting total radiation pattern and displays both quantities in separate figures as a function of the observation angle. Finally, the program is run for an array of four dipoles spaced $d = \lambda/4$ apart and operating at $f = 1.3$ GHz, for the four progressive phase shifts $\Phi = 0^\circ, 15^\circ, 30^\circ$ and 45° , so that the influence of the inter-element phase on the beam direction, beamwidth and side-lobe level can be examined and later compared with the array-factor calculator of CST Microwave Studio and with the full-wave CST model.

2. BACKGROUND

An antenna element, like a dipole, normally produces a wide radiation pattern, but for improved performance, more antenna elements are grouped together to produce a directional beam using an array of antennas. An array of antennas is made up of many identical antennas located in predetermined positions and excited at definite amplitudes and phases; this allows each individual field to be combined coherently to produce a narrow directional beam in the required direction. The concept of an array antenna is applied in radar and satellite communications, as well as mobile networks today, where 5G base station phased arrays and massive-MIMO arrays can steer beams electronically in the direction of individual subscribers [1].

The simplest type, which has been widely investigated, is called the uniform linear array (ULA), in which N identical antenna elements are arranged in a straight line. The geometry of the ULA can be characterized by three different variables, which include the number of elements N , the distance d between two adjacent elements, and the phase shift Φ between them. The electrical effect of the spacing is expressed through the wavenumber $k = 2\pi/\lambda$, where $\lambda = c/f$ is the operating wavelength. In this project the line on which the elements lie is taken as the array axis, the observation angle θ is measured from that axis, and the dipole elements extend in the direction perpendicular to it.

The radiation of an array is described by the principle of pattern multiplication. For an array of identical and identically oriented elements, the total far-field pattern is the product of two terms: the

element pattern, which is the field radiated by one isolated element, and the array factor (AF), which depends only on the geometry and excitation of the array and not on the type of element used:

$$E_{total}(\theta) = E_{element}(\theta) \times AF(\theta) \quad (1)$$

For a ULA of N elements with uniform amplitude excitation, constant spacing d and a progressive phase Φ between neighbouring elements, the field contributed by the n-th element differs from that of the first element by the total phase term $n\psi$, where ψ combines the spatial phase delay caused by the path-length difference between adjacent elements and the electrical phase imposed by the feed network:

$$\psi = k d \cos \theta + \Phi \quad (2)$$

Summing the contributions of all N elements gives the array factor, which is the quantity computed by the MATLAB code:

$$AF(\theta) = \sum e^{jn\psi}, \quad n = 0, 1, \dots, N-1 \quad (3)$$

This finite geometric series can be written in closed form, and its normalised magnitude is:

$$|AF(\theta)| = | \sin(N\psi/2) / \sin(\psi/2) | \quad (4)$$

The array factor reaches its maximum when $\psi = 0$, that is when $k d \cos \theta + \Phi = 0$. Solving for the beam-pointing angle gives

$$\cos \theta_{max} = - \Phi / (k d) \quad (5)$$

which shows directly that the main beam can be steered simply by changing the progressive phase Φ . For $\Phi = 0$ the maximum occurs at $\theta = 90^\circ$ (broadside radiation), whereas increasing Φ moves the beam away from broadside toward the array axis (the end-fire direction). With the values used in this project, $k d = (2\pi/\lambda)(\lambda/4) = \pi/2$, so each 15° increment of Φ produces a predictable shift of the beam direction, which is exactly the behaviour observed in the MATLAB and CST results.

3. DESIGN AND TEST PROCEDURES

The antenna used for this project is the pre-existing dipole antenna modelled with annealed copper from the first lab experiment has the parameters:

- The length of a pole $l=57.25\text{mm}$,
- The radius of the wire $r=0.5\text{mm}$,
- The gap between poles $g=1\text{mm}$.

After that new parameters are added to proceed for designing the array antenna.

- The distance between the consecutive array antennas $d=\lambda/4=57.69\dots\text{mm}$

3.1. MATLAB Design

To realise the formulation of the previous section, a MATLAB script was written that takes the array parameters as inputs and returns the array factor and the total radiation pattern. The constants are defined first: the number of elements $N = 4$, the frequency $f = 1.3$ GHz, the wavelength $\lambda = c/f \approx 0.2308$ m, the element spacing $d = \lambda/4$, and the wavenumber $k = 2\pi/\lambda$. The observation angle is swept over a full revolution using a fine grid of 3600 points between 0° and 360° , which guarantees that the nulls and side lobes are resolved accurately. The four progressive phase shifts of interest, $\Phi = 0^\circ, 15^\circ, 30^\circ$ and 45° , are stored in a list and processed inside a single loop.

For each phase shift the array factor is obtained directly from Equation (3) by accumulating the complex contribution of every element. In the code this is a short summation over the element index n , in which each term carries the phase $n(kd \cos\theta + \Phi)$:

```
AF = zeros(1, length(angles));
for n = 0:N-1
    AF = AF + exp(1j * n * (k*d*cos(angles) + phi));
end
AF = abs(AF);
```

The magnitude of the array factor is then normalised to its maximum and converted to a decibel scale with a floor of -40 dB so that the side-lobe structure is clearly visible. Because the dipole elements extend perpendicular to the array axis, their element pattern is essentially uniform in the plane analysed here; the element factor was therefore taken as constant in this cut, so the total pattern coincides with the normalised array factor (the small difference only becomes visible when the real dipole pattern is included in the CST model).

3.2. Applying Phase shift with Array Factor

Farfield Calculation of Antenna Arrays

☒ Activate farfield array

Array type: ☒ Rectangular ☐ Userdefined

Rectangular array settings

Elements in x, y, z: 4 1 1

Spaceshift in x, y, z: d 0 0

Phaseshift in x, y, z: 0 0 0

Update List

Antennas

No.	X	Y	Z	Amplitude	Phase
1	-86.5384...	0	0	1	0
2	-28.8461...	0	0	1	0
3	28.84615...	0	0	1	0
4	86.53846...	0	0	1	0

Import TSV... Import Array Task... Add... Modify... Delete

OK Cancel Help

Figure 1. The CST farfield array-factor calculator used to arrange four elements along the x-axis with spacing d and a progressive phase shift between consecutive elements.

After running the simulation, the antenna array is set from the **Array Factor** within the toolstrip on farfield window. 4 antennas on the x-axis with a distance d and a phaseshift in between consecutive antennas are arranged from this window. The phaseshift is set 0, 15, 30 and 45 degrees in order and results are collected.

3.3. Modelling Array Antenna

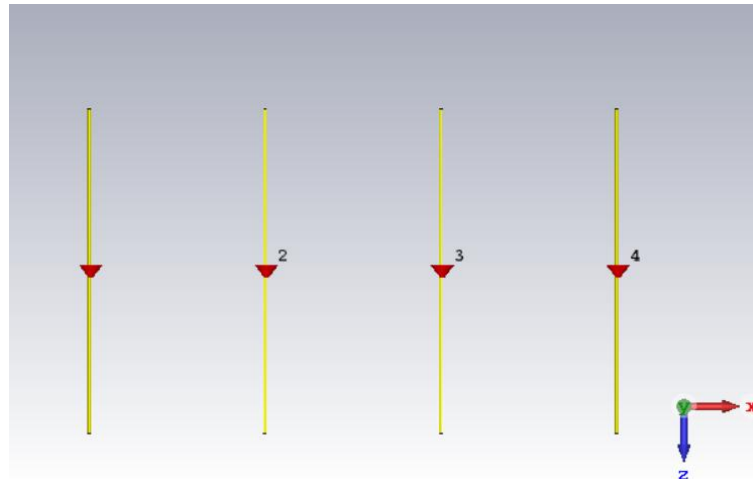


Figure 2. The full-wave CST model of the four-element dipole array, obtained by duplicating the single dipole and its port along the $+x$ direction.

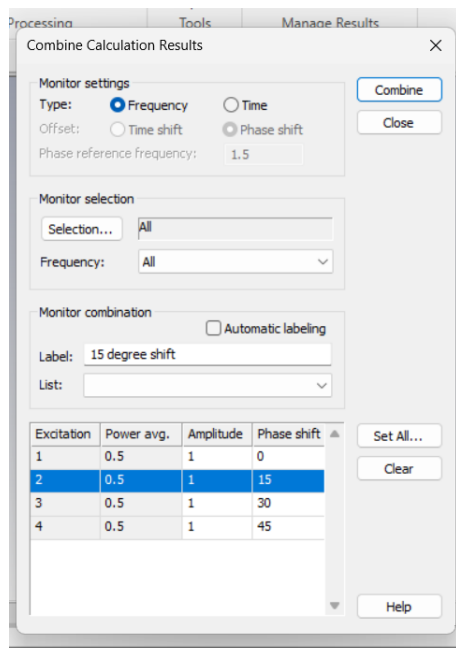


Figure 3. The Combine Results window used to superpose the four port excitations; a 15°

The array antenna modelling is completed by duplicating the antenna and the port in $+x$ direction with transform function in modelling window toolstrip.

The results are collected separately and combined by using the **Combine Results** function on the **View** window toolstrip. This is also where the phaseshifting is achieved. Figure 3 shows 15 degree shifting as an example. This operation also creates new s-parameters, combined farfield, e, h patterns and etc.

*progressive phase shift is applied here
as an example.*

4. RESULTS AND DISCUSSIONS

4.1. Farfield Patterns

4.1.1. MATLAB Results

For every phase shift the script produces a four-panel figure showing the array factor and the total pattern in both Cartesian (dB versus ϕ) and polar form, and it additionally reports the main-lobe direction, the half-power (3 dB) beamwidth and the side-lobe level. Three further figures overlay all four phase shifts so that the steering trend can be compared directly. The complete listing is not reproduced here; only its logic and the resulting figures are presented.

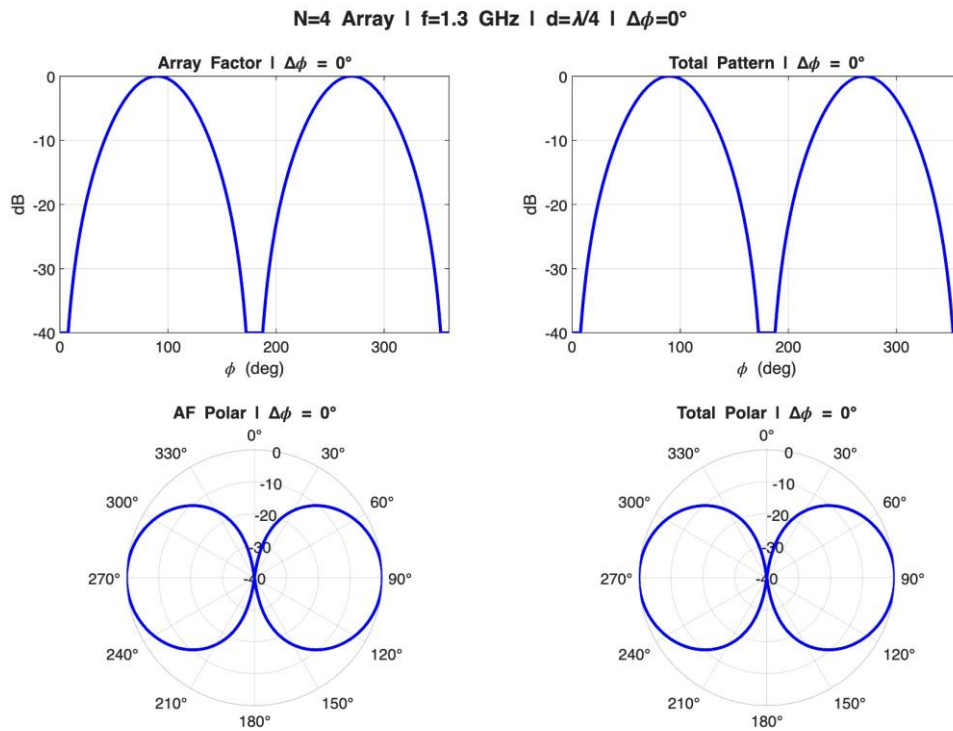


Figure 4. MATLAB results for $\Phi = 0^\circ$. The beam points to broadside ($\theta = 90^\circ$ and 270°) and the pattern is the symmetric two-lobe figure of a broadside array.

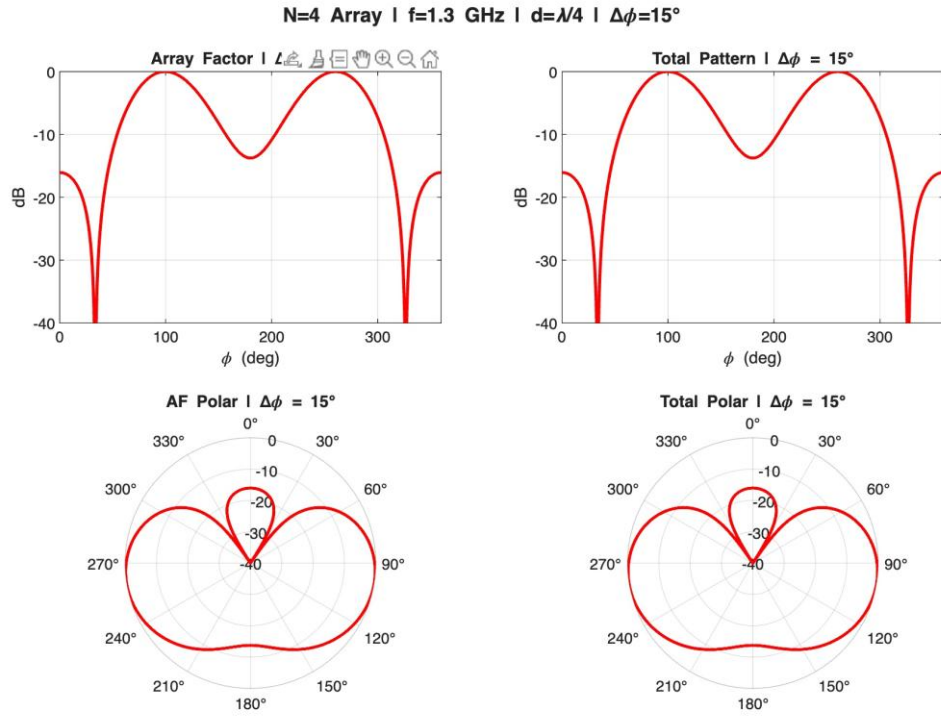


Figure 5. MATLAB results for $\Phi = 15^\circ$. The two beams shift slightly past broadside and a shallow back lobe begins to appear.

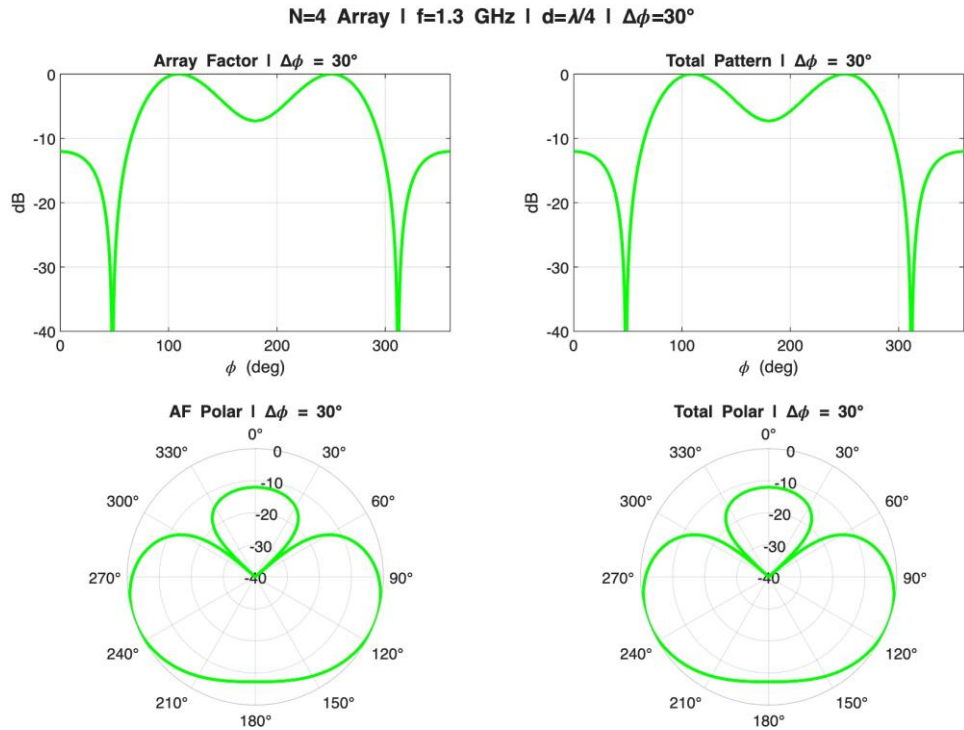


Figure 6. MATLAB results for $\Phi = 30^\circ$. The beams continue to tilt and broaden, and the dip between them rises to about -7 dB.

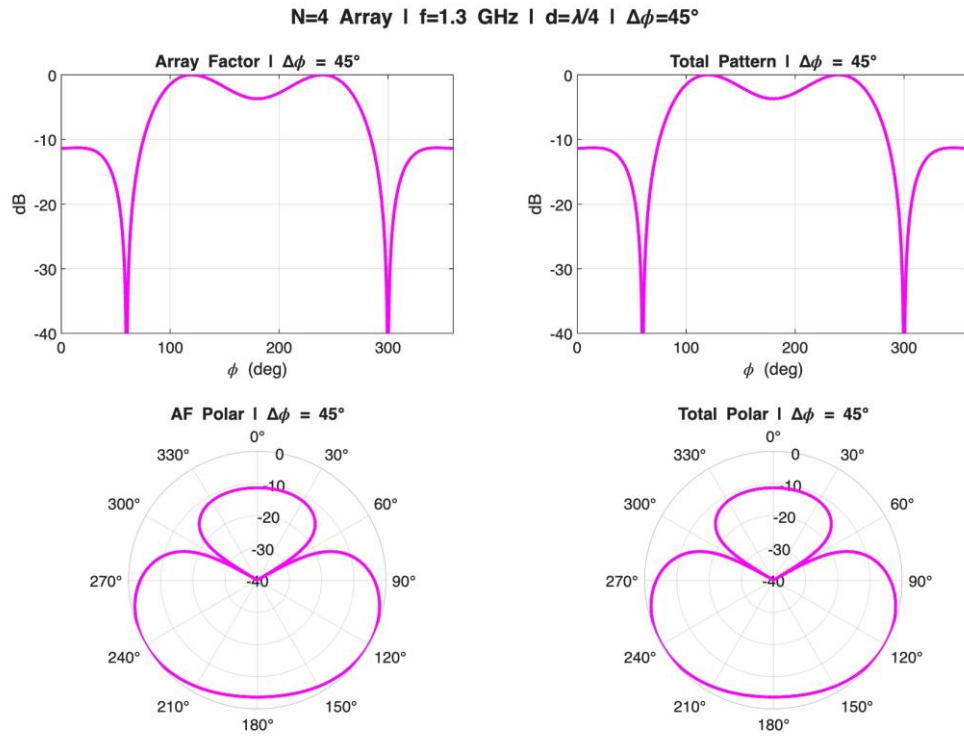


Figure 7. MATLAB results for $\Phi = 45^\circ$. The beams have steered to about 120° and 240° and begin to merge into a single broad lobe.

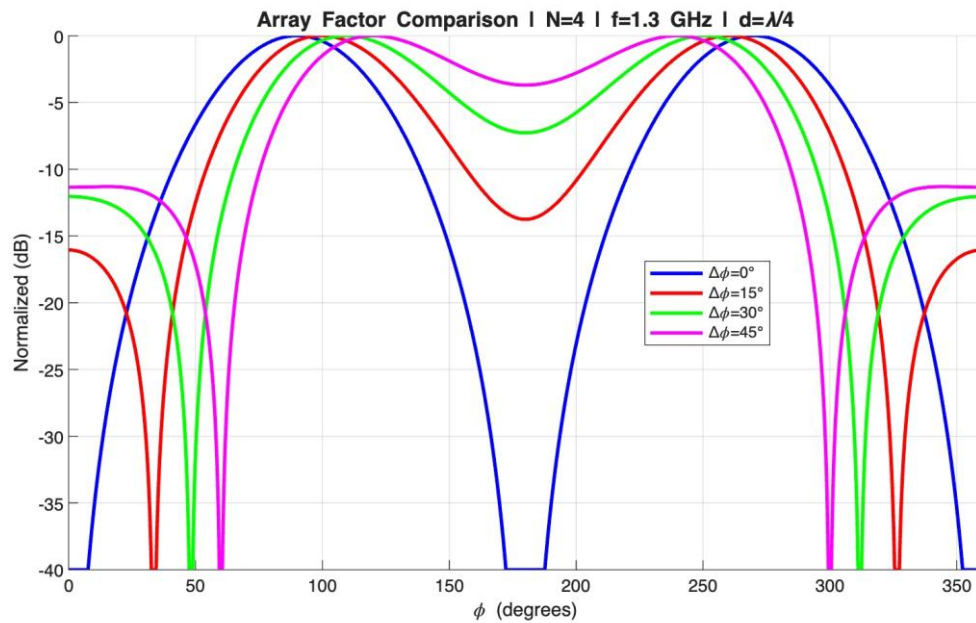


Figure 8. Array-factor comparison (Cartesian, dB) for the four phase shifts.

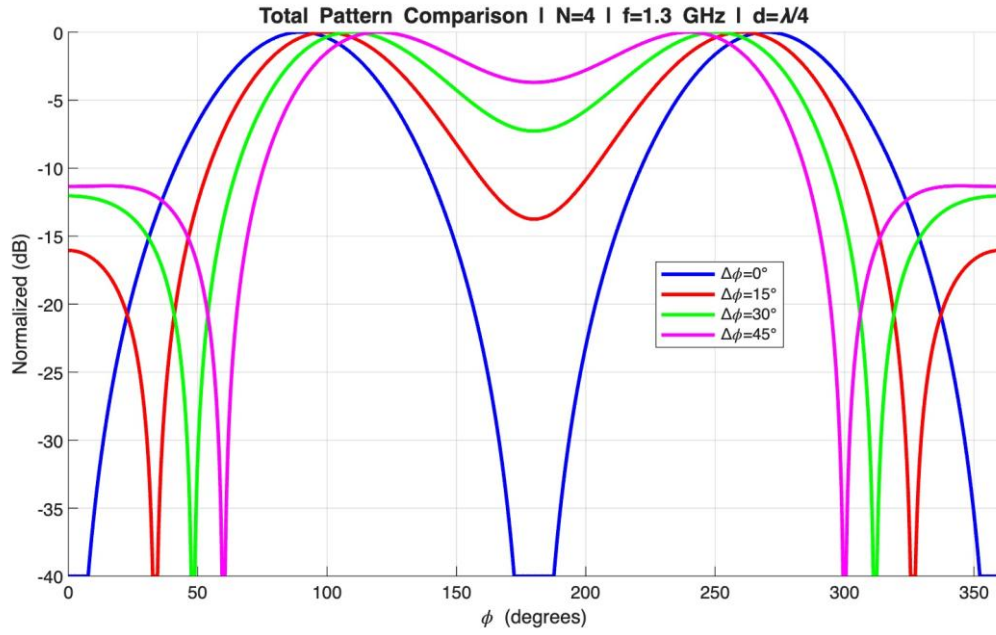


Figure 9. Total-pattern comparison (Cartesian, dB) for the four phase shifts. It is almost identical to Figure 8 because the element pattern is taken as uniform in this cut.

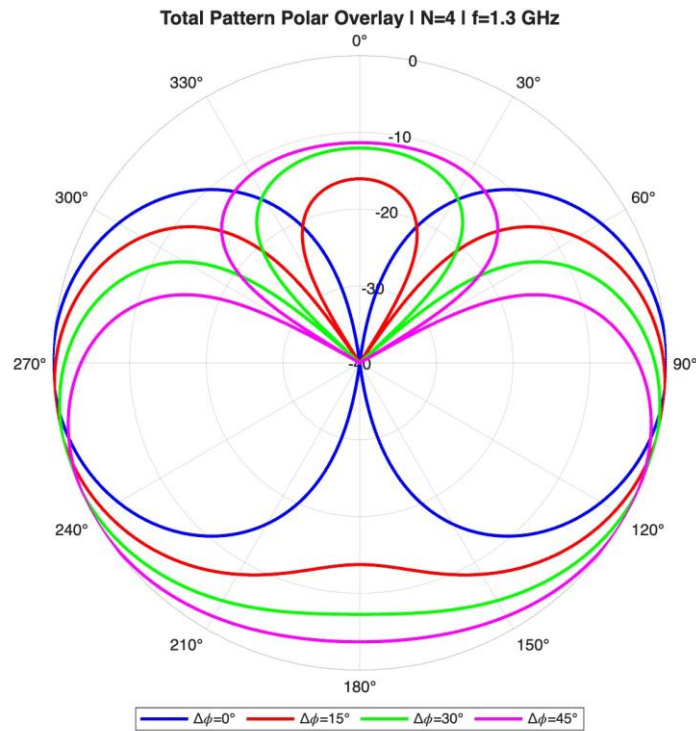


Figure 10. Polar overlay of the total pattern for the four phase shifts, showing the progressive steering of the main beam away from broadside.

To realise the formulation of the previous section, a MATLAB script was written that takes the array parameters as inputs and returns the array factor and the total radiation pattern. Constants are

introduced next: number of elements $N = 4$, frequency $f = 1.3$ GHz, wavelength $\lambda = c/f = 0.2308$ m, element spacing $d = \lambda/4$ and wavenumber $k = 2\pi/\lambda$. Angle of observation is covered in one full circle, utilizing a dense grid of 3600 points within 0° and 360° range that ensures a good resolution of nulls and side lobes. Four progressive phase shifts of interest are $\Phi = 0^\circ, 15^\circ, 30^\circ$ and 45° and are put in an array and processed in a single loop. The array factor for each phase shift is determined using Equation (3) by summing up the complex contribution of all elements. In the code this is a short summation over the element index n , in which each term carries the phase $n(kd \cos\theta + \Phi)$.

4.1.2. CST Results from Array Factor

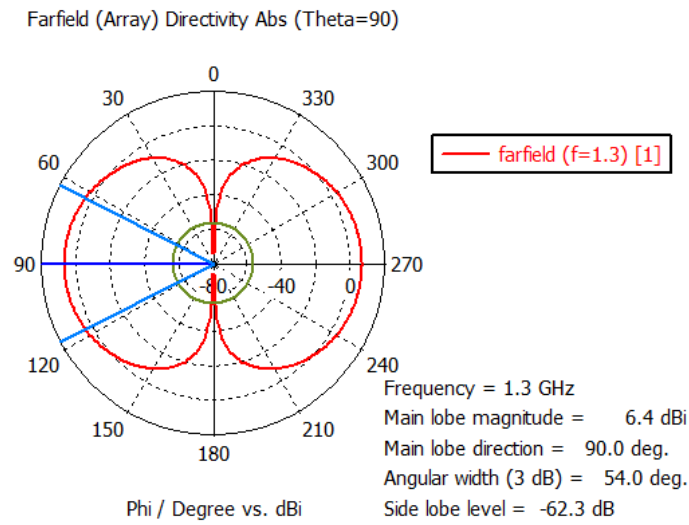


Figure 11. CST array-factor result for $\Phi = 0^\circ$. The main lobe is at 90° (broadside) with a 3 dB beamwidth of about 54° and a directivity of 6.4 dBi.

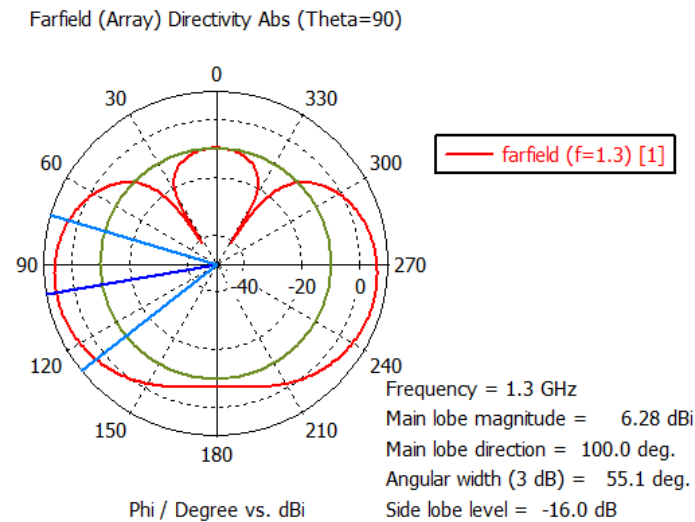


Figure 12. CST array-factor result for $\Phi = 15^\circ$. The main lobe steers to about 100° , with a 3 dB beamwidth near 55° and a directivity of 6.3 dBi.

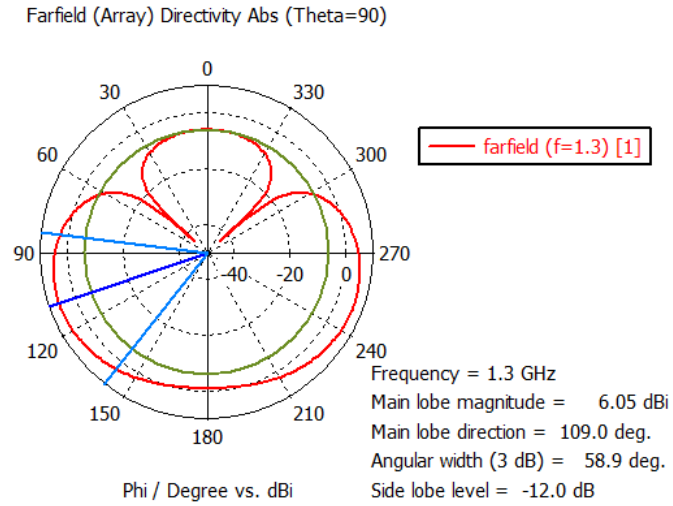


Figure 13. CST array-factor result for $\Phi = 30^\circ$. The main lobe steers to about 109° , with a 3 dB beamwidth near 59° .

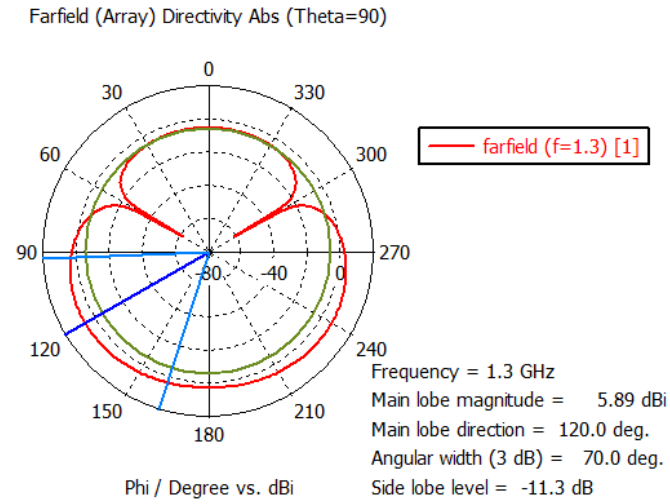


Figure 14. CST array-factor result for $\Phi = 45^\circ$. The main lobe steers to about 120° and the beam broadens to a 3 dB beamwidth of about 70° .

4.1.3. CST Results from modelling the array antenna

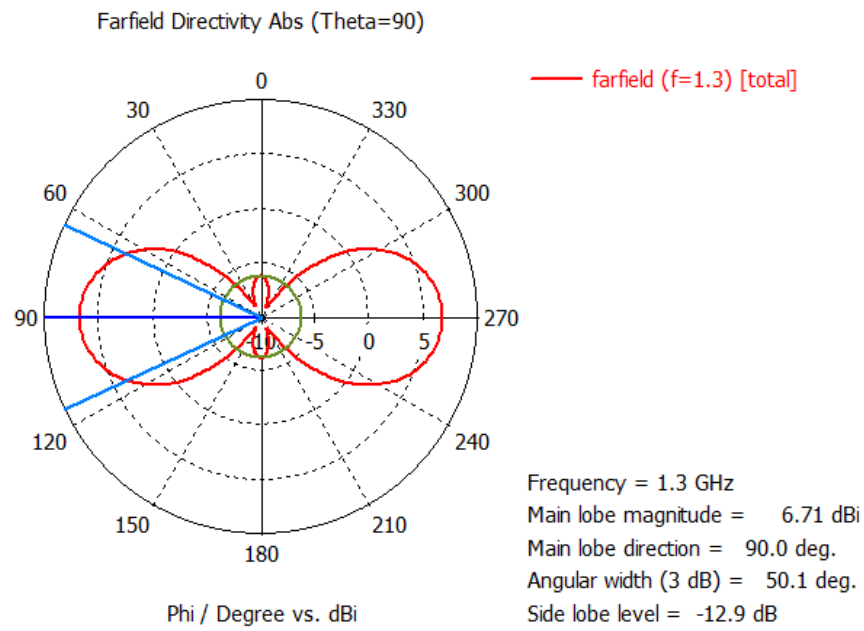


Figure 15. Full-wave CST model result for $\Phi = 0^\circ$. The combined far-field has its main lobe at 90° with a directivity of 6.7 dBi and a 3 dB beamwidth of about 50° .

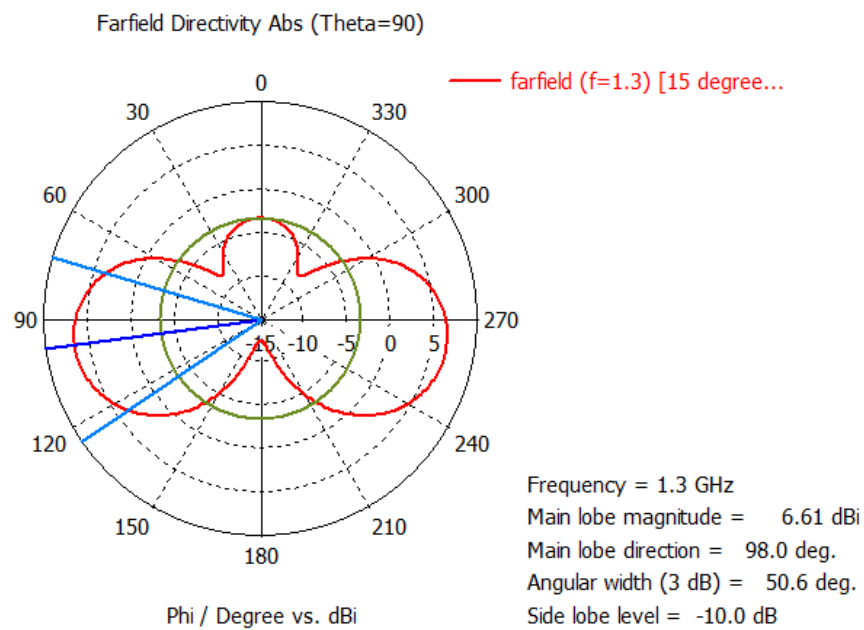


Figure 16. Full-wave CST model result for $\Phi = 15^\circ$. The main lobe steers to about 98° , close to the array-factor prediction.

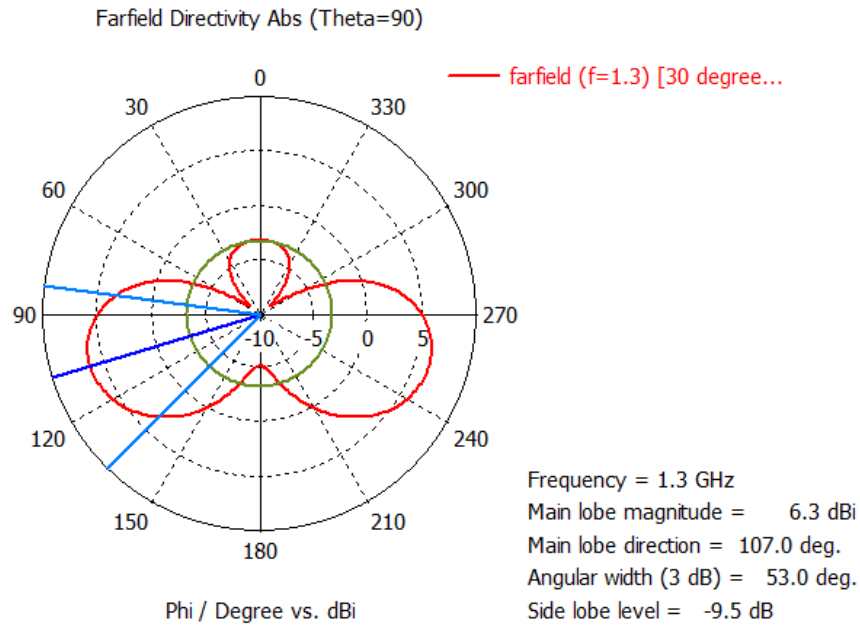


Figure 17. Full-wave CST model result for $\Phi = 30^\circ$. The main lobe steers to about 107° .

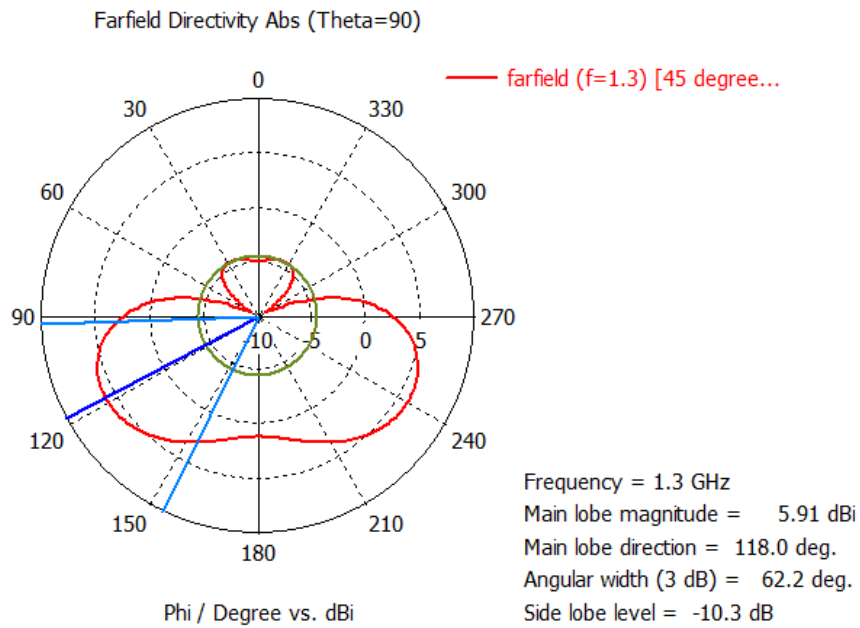


Figure 18. Full-wave CST model result for $\Phi = 45^\circ$. The main lobe steers to about 118° with a 3 dB beamwidth of about 62° .

4.2. S-Parameter Results

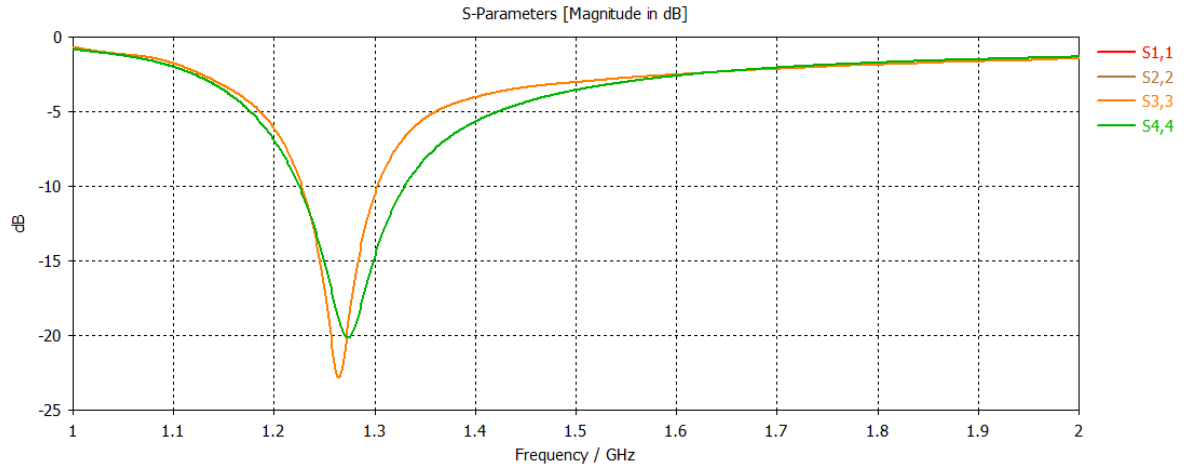


Figure 19. Simulated reflection coefficients ($S_{1,1}$ to $S_{4,4}$) of the four array ports versus frequency; all ports resonate near 1.3 GHz, where the return loss falls below -20 dB.

Reflection coefficients show that all four elements of the dipoles are at resonant frequency close to 1.3 GHz. Return losses remain below -20 dB, which means that there is impedance matching with no significant power loss between source and antenna elements. Similarities of S_{11} , S_{22} , S_{33} , and S_{44} further verify that elements of the array are identical in geometry and electromagnetic environment.

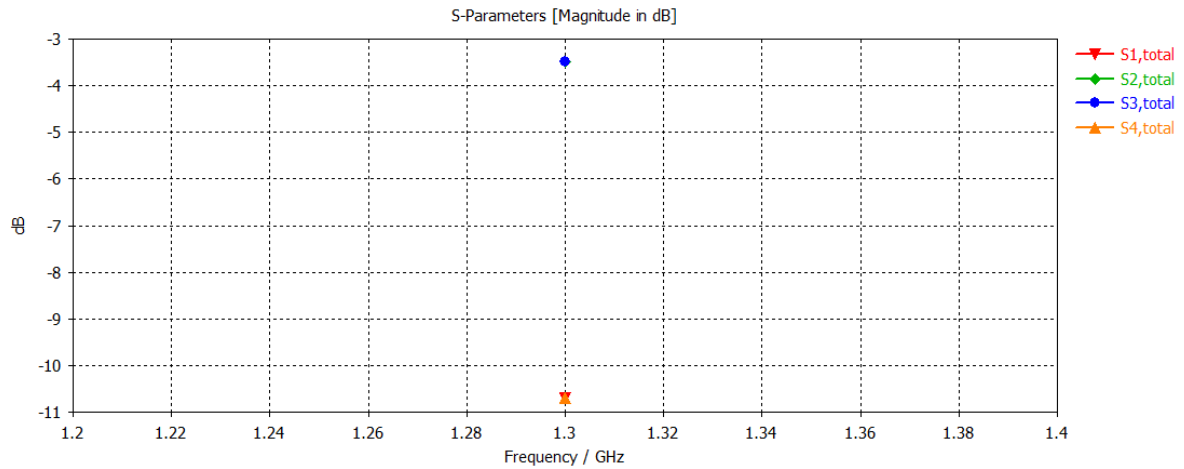


Figure 20. Combined reflection coefficients ($S_{1,total}$ to $S_{4,total}$) evaluated at the 1.3 GHz operating frequency.

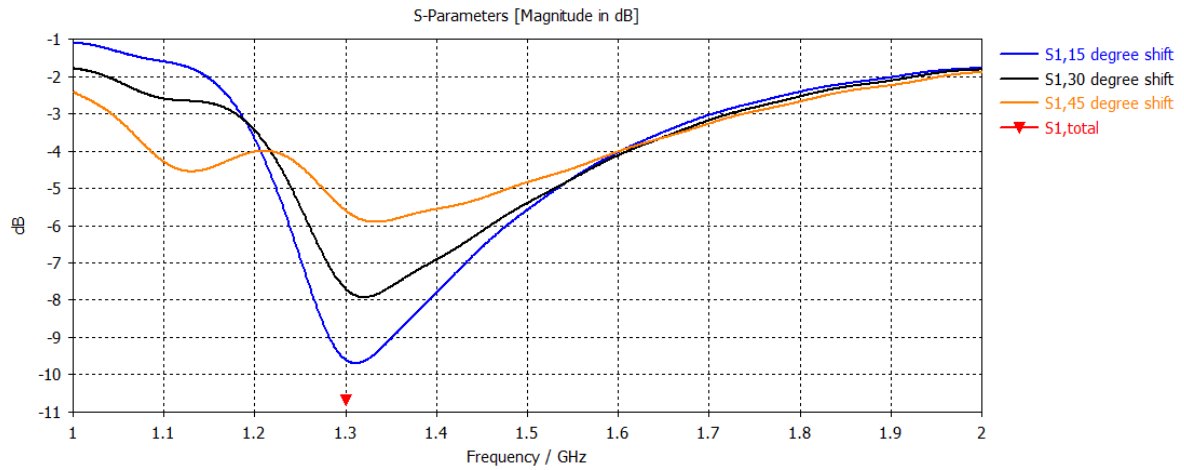


Figure 21. Reflection coefficient of port 1 ($S1$) for the 15° , 30° and 45° phase-shift cases, shown together with the combined $S1_{total}$.

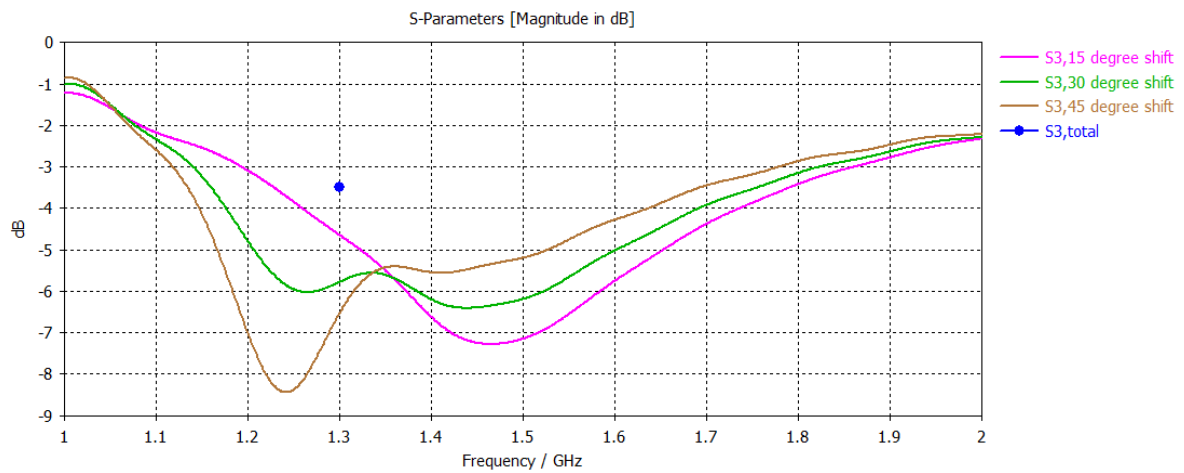


Figure 22. Reflection coefficient of port 3 ($S3$) for the 15° , 30° and 45° phase-shift cases, shown together with the combined $S3_{total}$.

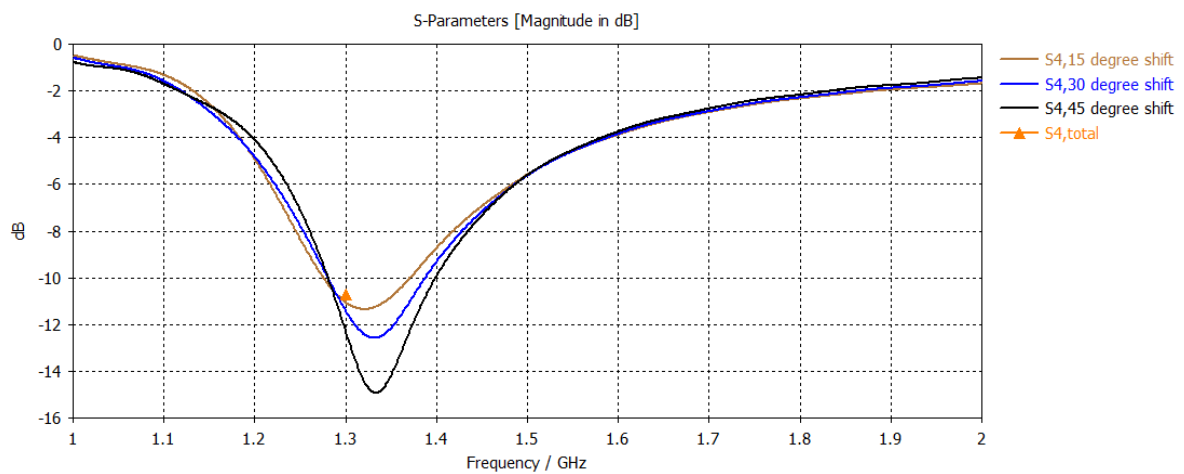


Figure 23. *Reflection coefficient of port 4 (S_4) for the 15° , 30° and 45° phase-shift cases, shown together with the combined S_4 , total.*

The phase-shifted excitation cases show only minor variations in the reflection coefficients. This behaviour is expected because changing the excitation phase primarily affects the direction of constructive interference in the far field rather than the impedance characteristics of the individual dipole elements. The observed differences arise mainly from mutual coupling effects and the interaction between neighbouring elements within the array.

4.3. Comparison of MATLAB, CST Array Factor and CST Full-Wave Results

The MATLAB analysis, CST Array Factor calculation method and the CST full-wave simulation method all demonstrate the same behaviour pattern: with the increase in the progressive phase shift between adjacent elements, the main beam is steered in the opposite direction compared to broadside radiation. When $\Phi = 0^\circ$, the beam is aimed at the angle of 90° , indicating broadside radiation. With the increase in phase shift up to 15° , 30° and 45° , the main beam is shifted to the angles of 100° , 109° and 120° , respectively.

MATLAB analysis and CST Array Factor calculation utilize the same formula; that is why their results differ very little. The minor differences may be attributed to the use of various plotting and normalizing procedures in the two methods.

In CST full-wave model, the same beam steering tendency can be seen, yet certain deviations from ideal predictions may occur. While the analytical array factor method only calculates the ideal radiation pattern, full-wave method takes into consideration the real sizes of the antenna elements, their excitation at ports and mutual coupling of neighboring antennas, which slightly shifts the beam direction and beam width.

Thus, the obtained similarity confirms the accuracy of both the analysis method and CST software implementation.

5. CONCLUSIONS

In this project, an array antenna consisting of four dipoles at 1.3 GHz was analysed via mathematical modelling, MATLAB software, and CST Microwave Studio simulation. Array factor for a linear array of antennas was derived and computed using MATLAB, thus enabling evaluation of radiation characteristics of the antenna with the progressive phase shifts being set to 0° , 15° , 30° , and 45° .

MATLAB analysis proved that an increase in progressive phase shift displaces the main lobe from the broadside and changes beamwidth and side-lobe characteristics

accordingly. The same pattern of behaviour was achieved in CST Array Factor simulation, which indicated that mathematical formulation is valid.

The full-wave CST simulation involved duplication of dipole elements and setting a corresponding phase excitation at the ports. Radiation patterns obtained were consistent with predictions made using MATLAB computations and CST Array Factor calculator. Discrepancies arose due to mutual coupling, physical dimension of the antenna and other real-world factors ignored in the analytical model.

S-parameter analysis proved that all the antennas have good return loss below -20 dB at the operational frequency of 1.3 GHz. Thus, one may conclude that the beam can be electronically steered without any mechanical movements of the antenna system.

Overall, the project was successful in proving the theory of uniform linear antennas as well as the connection that exists between excitation phase, array factor, and radiation performance.

6. REFERENCES

- [1] Balanis, C. A. (2016). *Antenna theory: Analysis and design* (4th ed.). Wiley.